Semester Project: Prediction of Dislocation Velocity in TaW Alloy Using ML

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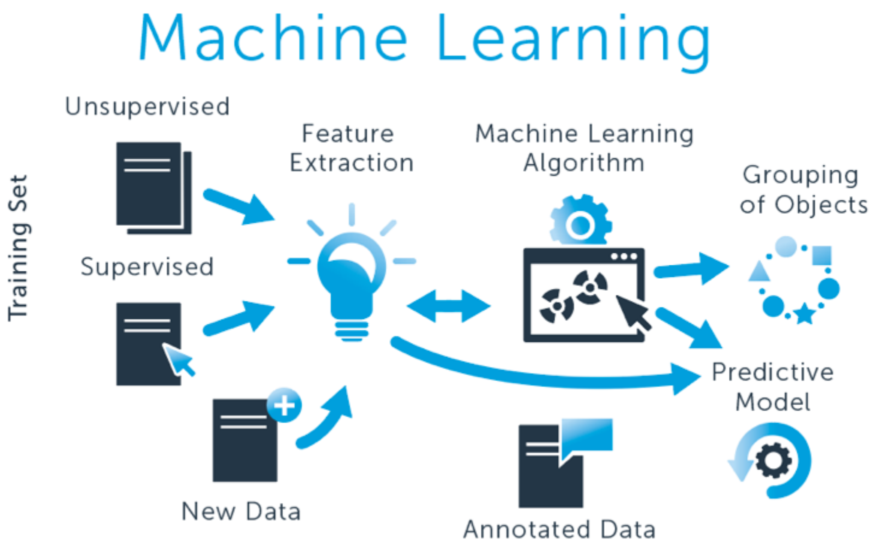

GitHub Repository link: https://github.com/vousoura/Final_Project_CMSE_802.git

Significance of the Problem

Understanding and predicting dislocation motion is critical to modeling plastic deformation in metals. Dislocation velocity, in particular, is a key factor influencing the strain rate, yield strength, and the overall mechanical response. Accurate predictions of dislocation velocity under different conditions—such as changes in solute concentration, slip system geometry, and temperature—are essential for the design of high-performance alloys. This project contributes to this goal by applying machine learning models to estimate dislocation velocity based on physically relevant features.

Scope and Limitations

This study focuses on the Ta-W alloy system and includes three predictive models: Least Squares Regression, Random Forest Regression, and XGBoost Regression. The model input features are solute concentration (Wt%), slip plane, temperature (K), and applied stress (MPa). While the dataset captures important variables affecting dislocation velocity, the study does not include other ivariables such as strain rate or microstructural features such as grain boundaries. Thus, the model's performance is therefore bounded by the quality and completeness of the provided data. Furthermore, all models are trained and tested on the same dataset, therefore no generalization can be applied to other alloy systems.

Project Implementation

This project aims to predict the dislocation velocity of TaW alloy using three machine learning models: Least Squares Regression, Random Forest Regression, and XGBoost Regression, and to compare their predictions. The target variable is the dislocation velocity, and the predictive features are solute concentration (Wt%), slip plane, temperature (K), and applied stress (MPa). The ultimate goal is to evaluate which model performs best and to interpret how each physical feature influences dislocation motion.

Data preprocessing was handled in a Python script `src/data_processing.py`. The raw data was loaded from `data/raw/raw_data.xlsx`, cleaned, and filtered to include only the three relevant slip planes: [110], [112], and [123]. The cleaned dataset was saved as `cleaned_data.csv` under `data/processed/`.

For each machine learning model, a separate Python script was written under the `src/` directory. These scripts load the cleaned dataset, split it into training and testing sets, train the model, and calculate the RMSE on the test set. Each trained model is saved as a `.pkl` file in the `models/` directory, and the RMSE result for each model is written to a `.txt` file in the `results/` folder.

In the `notebooks/` directory, individual Jupyter notebooks were used to analyze the model predictions. For each model, a set of plots was generated using Seaborn. These visualizations include actual vs predicted velocity, as well as how predicted velocity depends on each physical feature (solute concentration, slip plane, and temperature). A summary notebook also compares the RMSE results across the three models. Moreover, I was able to perform unit tests to all three models in the `tests/` directory where you can find all three `.py` files.

Through this workflow, I was able to compare the performance of each model and gain insights into which features most significantly affect the dislocation velocity. Among the three models, XGBoost provided the most accurate predictions and captured non-linear feature relationships, followed closely by the Random Forest model. The Least Squares model, while more interpretable, underperformed in terms of predictive accuracy and sensitivity to crystallographic features like slip plane which makes sense since it predicted linear relationships between each feature.

Interpretation of Visualizations

XGBoost Model

- **Actual vs Predicted Velocity**

  A strong linear trend indicates accurate model predictions.

- **Predicted Velocity vs Applied Stress**

Slight positive correlation: higher stress increases the predicted velocity, which aligns with expectations since applied stress governs plastic deformation.

- **Predicted Velocity vs Solute Concentration**

  Slight negative trend: increased solute concentration slightly reduces velocity. This is expected, as solutes hinder dislocation motion.

- **Predicted Velocity vs Slip Plane**

  Small variation observed: slip planes 110 and 112 show higher predicted velocity. This is interesting given the known changes in slip behavior across planes in BCC structures.

- **Predicted Velocity vs Temperature**

  Slight negative trend: velocity decreases slightly with increasing temperature due to phonon scattering and the thermally activated kink-pair mechanism.

- **Velocity Distribution**

  The histogram shows the velocity distribution is centered around 13–15 m/s.

- **Box Plot: Velocity by Solute Concentration**

  Shows a slight decrease in predicted velocity with increasing solute concentration, which aligns with physical expectations.

Random Forest Model

- **Actual vs Predicted Velocity:**

  This plot shows a strong alignment between actual and predicted velocity values. Most points fall close to the diagonal line, suggesting the Random Forest model is performing well overall in predicting dislocation velocity. This is an important baseline for assessing model accuracy.

- **Predicted Velocity vs Applied Stress:**

  A positive correlation is visible: as applied stress increases, the model predicts higher dislocation velocities. This follows physical intuition — more stress promotes faster

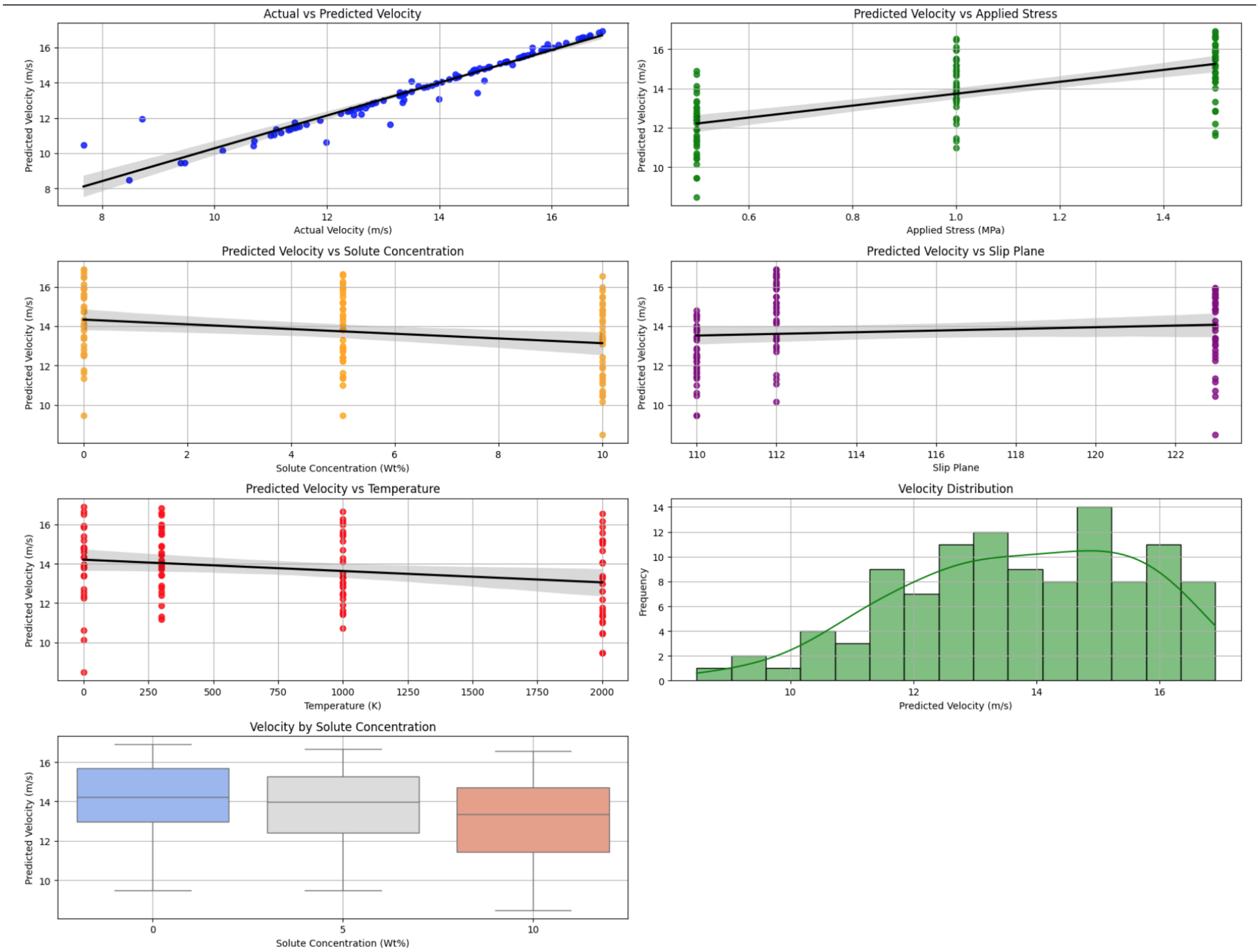

Figure 1: **Predicted Dislocation Velocity Analysis using the XGBoost Model.**

dislocation motion. However, since applied stress is not your main variable of interest, it's used more as a control feature here.

- **Predicted Velocity vs Solute Concentration:**
  This plot reveals a slight negative trend, where higher solute concentrations lead to lower predicted dislocation velocities. The Random Forest model appears to capture the solute strengthening effect — where solute atoms hinder dislocation motion. However, the relationship isn't very sharp, which may reflect the limitations of the data distribution or the model's sensitivity.

- **Predicted Velocity vs Slip Plane:**
  This figure shows predicted velocities across the three slip planes (110, 112, 123). The variation between slip planes is subtle, but the model does assign slightly different velocity values depending on the plane. This suggests that the Random Forest model recognizes some influence of crystallographic slip system on dislocation velocity.

- **Predicted Velocity vs Temperature:**
  There is a mild downward trend: the model predicts that dislocation velocity slightly decreases as temperature increases. This might reflect drag effects at higher temperatures, or possibly noise in the data. The trend is weak but consistent, showing that the model is at least partially sensitive to temperature effects.

- **Velocity Distribution (Histogram):**
  The histogram shows the distribution of predicted dislocation velocities. It is slightly right-skewed and centered around 13–15 m/s. This plot helps verify that the model isn't producing outliers or extreme values, and that its predictions follow a physically plausible distribution.

- **Box Plot: Velocity by Solute Concentration:**
  This final plot groups predicted velocities by solute concentration values. You can see a slight decrease in median velocity as concentration increases ($0 \rightarrow 5 \rightarrow 10$ wt). This aligns with your hypothesis: higher solute concentrations slow down dislocations. The

spread also increases at higher concentrations, suggesting more variability in the model's predictions at those levels.

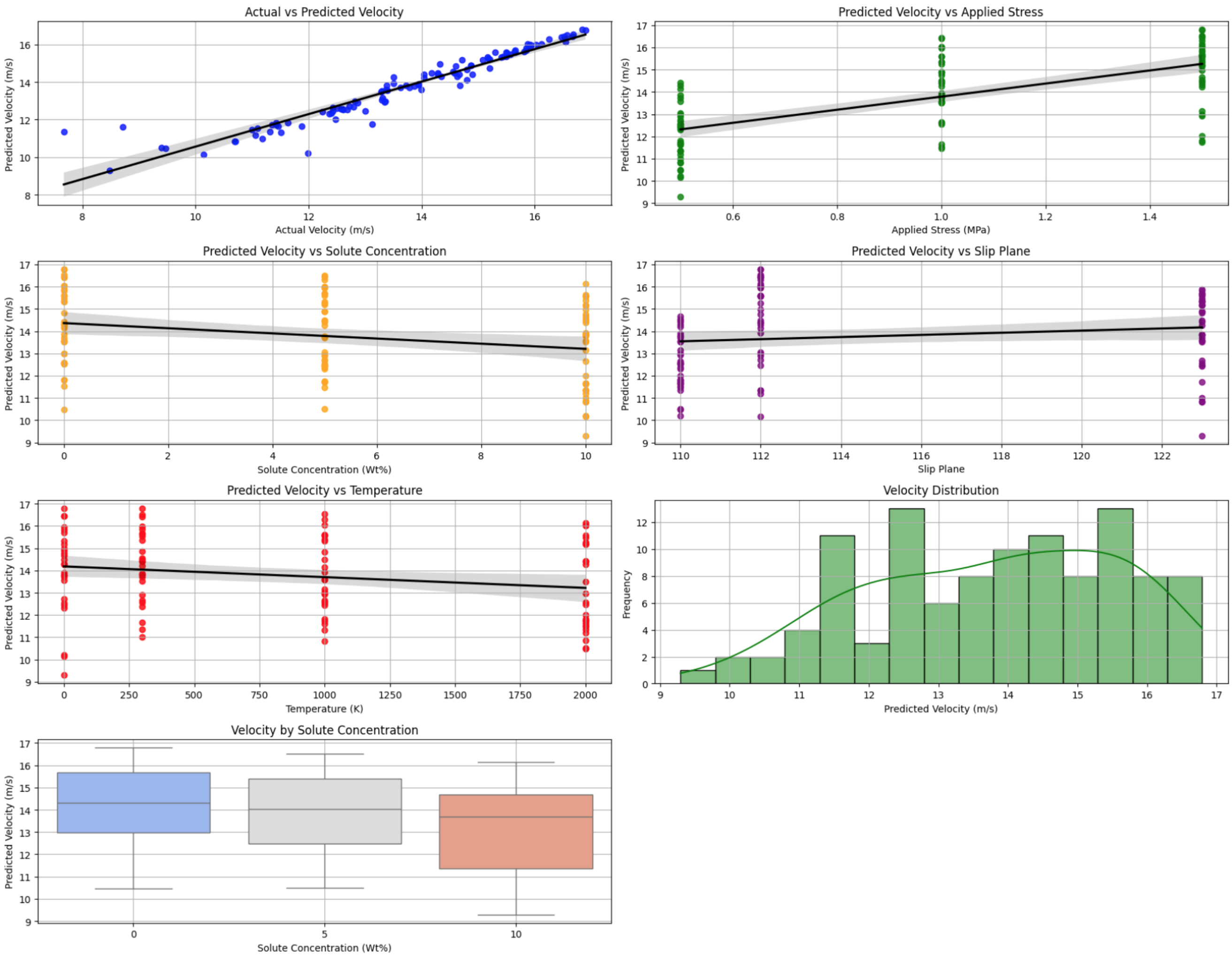

Figure 2: **Predicted Dislocation Velocity Analysis using the Random Forest Model.**

Least Squares Model

- **Actual vs Predicted Velocity:**

  This plot shows a generally upward trend, indicating that the model captures the basic relationship between features and velocity. However, compared to the Random Forest and XGBoost models, the points are more spread out around the regression line, which suggests lower accuracy and more prediction error.

- **Predicted Velocity vs Applied Stress:**

  There is a clear positive linear trend: as applied stress increases, the predicted velocity also increases. This aligns with expectations from physical theory, but the trend is smoother and more constrained due to the linear nature of the model.

- **Predicted Velocity vs Solute Concentration:**

  A visible negative slope is present — predicted velocity decreases as solute concentration increases. This means the Least Squares model correctly identifies the solute strengthening effect, where higher concentrations of solute atoms slow down dislocation motion.

- **Predicted Velocity vs Slip Plane:**

  Here, the model shows very limited variation across the slip planes (110, 112, 123). The line is nearly flat, meaning that Least Squares treats all slip planes similarly and fails to capture subtle differences in their effect on velocity.

- **Predicted Velocity vs Temperature:**

  This plot shows a small but steady decline in velocity with increasing temperature, indicating a weak negative correlation. This trend is consistent with physical expectations but not very pronounced in this model.

- **Velocity Distribution (Histogram):**

  The histogram is more symmetric and centered around 13.5–14.5 m/s. There are no major outliers, but the range is slightly narrower compared to Random Forest and XGBoost, suggesting the model outputs less variation in predictions.

- **Box Plot: Velocity by Solute Concentration:**

  As with the Random Forest model, the median predicted velocity decreases with increasing solute concentration ($0 \rightarrow 5 \rightarrow 10$ wt). The Least Squares model appears to capture this trend quite clearly, but the spread of values is more compressed, reflecting its limited capacity to model non-linearity.

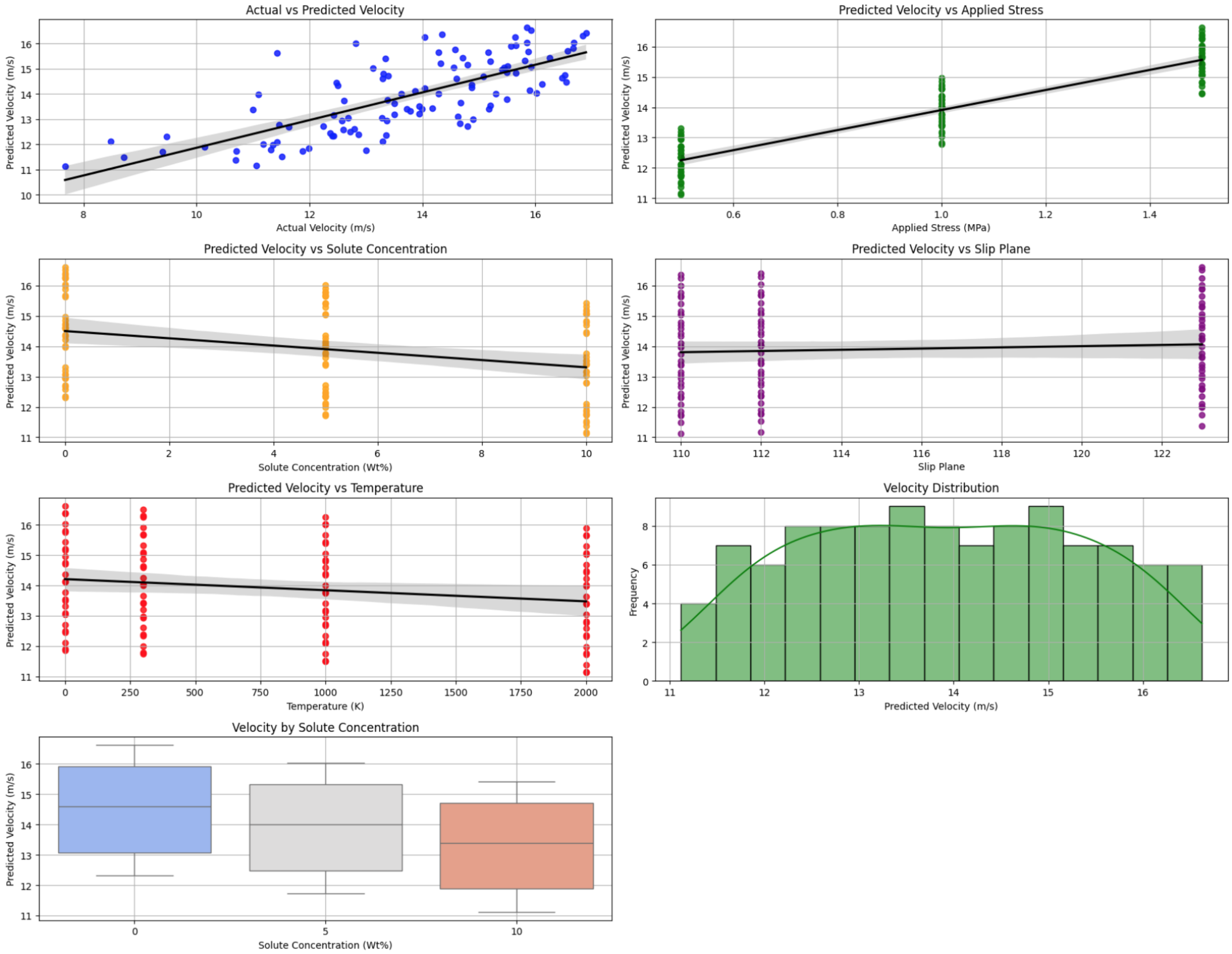

Figure 3: **Predicted Dislocation Velocity Analysis using the Least Squares Model.**

Model Comparison Summary

Overall, the XGBoost and Random Forest models both demonstrated strong predictive performance, with the XGBoost model slightly outperforming the others in accuracy and consistency. Both tree-based models captured the expected physical trends: dislocation velocity decreased with increasing solute concentration and temperature, and varied slightly across slip planes — with slip planes 110 and 112 typically yielding higher predicted velocities. XGBoost produced better actual vs. predicted velocity alignment, indicating strong model generalization, while Random Forest showed slightly more spread and variability.

In contrast, the Least Squares model was able to capture basic linear relationships, such as the decrease in velocity with higher solute concentration and temperature. However, it struggled to model non-linear interactions, especially between velocity and slip plane. The predictions were more constrained, and the velocity distribution was narrower compared to the tree-based models.

In conclusion, XGBoost provided the most robust and physically interpretable predictions across all key features, making it the most suitable model for predicting dislocation velocity in this dataset. Random Forest was a close second, while Least Squares, although useful for understanding linear trends, was limited in flexibility and predictive accuracy.

Conclusion

Summary of Achievements

In this project, I successfully implemented three machine learning models—Least Squares Regression, Random Forest Regression, and XGBoost Regression—to predict dislocation velocity based on solute concentration, slip plane, temperature, and applied stress. I developed a complete data pre-processing pipeline, trained and evaluated each model, and visualized the results using a consistent set of plots. All code was modularized, documented, and organized in a reproducible directory structure. Among the three models, XGBoost achieved the lowest RMSE, demonstrating the strongest predictive performance.

Evaluation of Approach

The machine learning approach proved effective in capturing the complex dependencies between physical features and dislocation velocity. Tree-based models (XGBoost and Random Forest) performed better than the linear Least Squares model, highlighting the importance of non-linear relationships in the data. The use of consistent visualization techniques across models enabled a fair comparison and supported physical interpretation of model predictions. Limitations of this approach include the narrow scope of input features and the reliance on a fixed dataset, which may limit generalization to other systems or microstructural conditions.

Future Implications

As a natural extension of this work, which is beyond the scope of this class, I plan to perform molecular dynamics (MD) simulations using LAMMPS to directly simulate dislocation dynamics in BCC metals. These simulations will provide atomic-scale insights into how solute atoms, slip systems, and thermal effects influence dislocation motion. The goal is to compare MD-derived dislocation velocities with ab initio calculations, validate model assumptions, and potentially generate new training data for future improvements in model accuracy and physical interpretability.

References

1. Kedharnath, A., Kapoor, R., and Sarkar, A. (2025). *Understanding dislocation velocity in TaW using explainable machine learning*. **Tungsten**, 7, 327–336. https://doi.org/10.1007/s42864-024-00306-9

2. GitHub Repository (data and supplemental code): https://github.com/kedhar1992/dislocation-mobility/